

4.2 Annihilators

Section 1 built the dual space V^* , the space of linear measurements on V . Now we connect subspaces of V with subspaces of V^* . A plane in $\mathbb{R}^3$ can be described by a spanning list or by an equation, and an equation is a functional. This section collects all the equations that a set of vectors satisfies into one subspace of V^* , the annihilator, counts its dimension, and shows that it turns sums into intersections and intersections into sums.

4.2.1 The equations a subspace satisfies

Every object comes with its subobjects. The dual space V^* has subspaces of its own, and the question is which of them are attached to subspaces of V . The standard example shows the way.

Let $U = \text{span}((1, 0, 1), (0, 1, 1))$, a plane in $\mathbb{R}^3$. The functional $\varphi(x, y, z) = x + y - z$ vanishes on both spanning vectors, so by linearity it vanishes on all of U : the plane satisfies the equation $x + y - z = 0$. So do $2x + 2y - 2z = 0$ and the trivial equation $0 = 0$. Which functionals vanish on U ? By Theorem 4.1.1 a functional is $\psi(x, y, z) = ax + by + cz$. It vanishes on the two spanning vectors if and only if $a + c = 0$ and $b + c = 0$, that is, $\psi = -c(x + y - z) = -c\varphi$. So the equations of the plane form the **line** $\text{span}(\varphi)$ in $(\mathbb{R}^3)^*$.

For the z -axis $\text{span}((0, 0, 1))$, the same computation gives $c = 0$: the equations are $ax + by = 0$, a **plane** in $(\mathbb{R}^3)^*$. A bigger subspace satisfies fewer equations, and in both cases the dimensions add up to 3. The set of equations deserves a name, and the pattern deserves a theorem.

The annihilator of a set of vectors is the space of all linear equations that every one of those vectors satisfies.

Definition 4.2.1 — Annihilator

Let V be a vector space over F , and let $S \subseteq V$ be **any subset**. The **annihilator** of S is

$$S^0 := \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}.$$

In words: S^0 is a subset of the **dual space** V^* , not of V . A functional belongs to it when it vanishes at **every** vector of S ; a single non-zero value disqualifies it. The set S needs no structure: it may be a subspace, a finite list of vectors, or even empty. The superscript 0 recalls the value that the functionals take on S .

Well-definedness. The definition makes sense for every subset. What needs checking is that the result is a subspace, which is what makes its dimension meaningful.

Proposition 4.2.1 — The Annihilator Is a Subspace

Let V be a vector space over F , and let $S \subseteq V$. Then S^0 is a subspace of V^* .

Proof. We use the subspace test (Theorem 1.2.1) in V^* . The zero functional sends every $\mathbf{s} \in S$ to 0, so it lies in S^0 . Let $\varphi, \psi \in S^0$ and $c \in F$. For every $\mathbf{s} \in S$, the pointwise

operations of Definition 4.1.2 give

$$(\varphi + \psi)(\mathbf{s}) = \varphi(\mathbf{s}) + \psi(\mathbf{s}) = 0 + 0 = 0, \quad (c\varphi)(\mathbf{s}) = c\varphi(\mathbf{s}) = c \cdot 0 = 0.$$

Hence $\varphi + \psi \in S^0$ and $c\varphi \in S^0$, and S^0 is a subspace of V^* . ■

Example. (First Annihilators)

- (a) For any vector space V , find $\{\mathbf{0}\}^0$, $\emptyset^0$ and V^0 .
- (b) In $M_2(F)$, find the annihilator of the single matrix E_{11} .
- (c) In $\mathbb{R}[x]_{\leq 2}$, find the annihilator of $S = \{1, x\}$.

Solution.

- (a) Every functional sends $\mathbf{0}$ to 0 (Theorem 3.1.1), so $\{\mathbf{0}\}^0 = V^*$. The condition “for all $\mathbf{s} \in \emptyset$ ” holds vacuously, so $\emptyset^0 = V^*$. A functional in V^0 takes the value 0 at every vector, so it is the zero functional, and $V^0 = \{0\}$. These degenerate cases fix the direction of everything that follows: the smallest subspace of V has the largest annihilator, and the largest has the smallest.
- (b) Let $\varepsilon_{ij}(A) = a_{ij}$ be the dual basis of the standard basis $(E_{11}, E_{12}, E_{21}, E_{22})$. By Theorem 4.1.3 (c), every $\psi \in M_2(F)^*$ is $\psi = \sum_{i,j} \psi(E_{ij}) \varepsilon_{ij}$. Such a ψ lies in $\{E_{11}\}^0$ exactly when $\psi(E_{11}) = 0$. Hence

$$\{E_{11}\}^0 = \text{span}(\varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}),$$

the functionals $A \mapsto c_{12}a_{12} + c_{21}a_{21} + c_{22}a_{22}$. It has dimension 3, and $3 + 1 = 4 = \dim M_2(F)$.

- (c) Let $(\varphi_0, \varphi_1, \varphi_2)$ be the dual basis of $(1, x, x^2)$, so φ_k reads off the coefficient of x^k . A functional $\psi = \psi(1)\varphi_0 + \psi(x)\varphi_1 + \psi(x^2)\varphi_2$ lies in S^0 exactly when $\psi(1) = \psi(x) = 0$, that is, when $\psi = \psi(x^2)\varphi_2$. So $S^0 = \text{span}(\varphi_2)$: the only equations satisfied by both 1 and x are the multiples of “the coefficient of x^2 is 0”.

Non-example by minimal change. Shift the plane of the opening example off the origin: $A = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 1\}$. The set is still cut out by the same left-hand side, but $\varphi(x, y, z) = x + y - z$ takes the value 1, not 0, on A , so the defining clause $\varphi(\mathbf{s}) = 0$ fails and $\varphi \notin A^0$. In fact $A^0 = \{0\}$: the set A contains $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, -1)$, and a functional $ax + by + cz$ vanishing at these three vectors has $a = b = c = 0$. Annihilators record only **homogeneous** equations.

Why this definition. We allow any subset S , not only subspaces, because subspaces are usually handed to us by spanning lists, and we want to test a functional on the list alone. The theorem at the end of this section shows that this loses nothing: $S^0 = (\text{span } S)^0$. The name is literal. To annihilate is to reduce to nothing, and the functionals in S^0 reduce every vector of S to 0.

Warning. U^0 lives in V^* , and it is not a complement of U . It is tempting to turn the functional $ax + by$ into the vector (a, b) and to picture U^0 as a subspace of F^2 “perpendicular” to U . Over $\mathbb{F}_2$ this picture fails completely. Let $U = \text{span}((1, 1)) \subseteq \mathbb{F}_2^2$.

A functional $ax + by$ vanishes on $(1, 1)$ exactly when $a + b = 0$, that is, $a = -b$, so $U^0 = \{0, x + y\}$. The functional $x + y$ corresponds to the vector $(1, 1)$, which lies **in** U . So the “picture” of U^0 is U itself, not a complement. Perpendicularity needs an inner product, which is the orthogonal complement of Chapter 10; the annihilator needs nothing.

The payoff of the definition is the pattern we saw in the opening example. The equations of a subspace take up exactly the dimensions that the subspace leaves free.

4.2.2 Counting equations

Theorem 4.2.2 — Dimension of the Annihilator

Let V be a finite-dimensional vector space over F , and let U be a subspace of V . Then

$$\dim U + \dim U^0 = \dim V.$$

More precisely, if $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ is a basis of V extending a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of U , with dual basis $(\varphi_1, \dots, \varphi_n)$, then $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0 .

Idea. This is a dimension formula, so we start from the smallest subspace, U , take a basis and extend it to V . The dual basis now splits in two. The last $n - m$ functionals vanish on every $\mathbf{u}_i$, by the delta, so they vanish on U . The first m do not. To see that nothing else is in U^0 , expand a functional $\psi \in U^0$ in the dual basis: its coefficients are its values on the basis, and the first m values are $\psi(\mathbf{u}_i) = 0$.

Proof. Since U is a subspace of the finite-dimensional space V , it has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ (Theorem 1.6.7, Corollary 1.5.5), and by the Basis Extension Theorem (Theorem 1.6.5) it extends to a basis $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ of V , where $n = \dim V$. Let $(\varphi_1, \dots, \varphi_n)$ be the dual basis of $\mathcal{B}$ (Theorem 4.1.3).

The φ_j with $j > m$ lie in U^0 . Let $j > m$ and $\mathbf{u} \in U$, say $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m$. By Theorem 3.1.4, $\varphi_j(\mathbf{u}) = \sum_{i=1}^m a_i\varphi_j(\mathbf{u}_i) = 0$, since $\varphi_j(\mathbf{u}_i) = \delta_{ji} = 0$ for $i \leq m < j$.

They span U^0 . Let $\psi \in U^0$. By Theorem 4.1.3 (c),

$$\psi = \sum_{i=1}^m \psi(\mathbf{u}_i) \varphi_i + \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j = \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j,$$

because $\psi(\mathbf{u}_i) = 0$ for $\mathbf{u}_i \in U$. Hence $U^0 \subseteq \text{span}(\varphi_{m+1}, \dots, \varphi_n)$, and the reverse inclusion holds because U^0 is a subspace (Proposition 4.2.1, Theorem 1.3.1).

They are independent, being part of the basis $(\varphi_1, \dots, \varphi_n)$ of V^* . Therefore $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0 , and $\dim U^0 = n - m = \dim V - \dim U$. This proves the theorem. ■

Alternatively, restrict functionals to U . The map $R: V^* \rightarrow U^*$, $\varphi \mapsto \varphi|_U$, is linear, and its kernel is U^0 by definition. It is surjective: given $\xi \in U^*$, Theorem 3.1.5 gives $\varphi \in V^*$ with $\varphi(\mathbf{u}_i) = \xi(\mathbf{u}_i)$ and $\varphi(\mathbf{v}_j) = 0$, and then $\varphi|_U = \xi$ because both agree on a basis of U . By the Rank–

Nullity Theorem (Theorem 3.3.1) and Corollary 4.1.4, $\dim V = \dim V^* = \dim U^0 + \dim U^* = \dim U^0 + \dim U$. This route shows **why** the formula holds: a functional on V is a functional on U plus a choice of values off U , and U^0 measures that choice.

The theorem is the second signature move of the chapter: **pass to the dual and count**. One immediate consequence: if U is a **proper** subspace of a finite-dimensional V , then $\dim U^0 > 0$, so some **non-zero** functional vanishes on U . Equivalently, a list spans a finite-dimensional V if and only if the only functional that kills every vector of the list is 0: apply the previous sentence to the span U of the list, and note that a functional kills every vector of the list exactly when it kills U , by Theorem 3.1.4.

In F^n the whole computation becomes elimination, because functionals are rows.

Example. (An Annihilator as a Null Space)

Find a basis of U^0 for $U = \text{span}((1, 2, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$, and check the dimension formula.

Solution. By Theorem 4.1.1, every functional is $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$ for a unique $\mathbf{a} = (a_1, a_2, a_3)$. By linearity, $\varphi_{\mathbf{a}}$ vanishes on U if and only if it vanishes on the two spanning vectors (a combination of vectors killed by $\varphi_{\mathbf{a}}$ is killed, by Theorem 3.1.4). These two conditions are

$$a_1 + 2a_2 = 0, \quad a_2 + a_3 = 0, \quad \text{that is,} \quad A\mathbf{a} = \mathbf{0} \quad \text{with} \quad A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

the matrix whose **rows** are the spanning vectors. So $\varphi_{\mathbf{a}} \in U^0$ if and only if $\mathbf{a} \in \text{nul}(A)$. Subtracting twice row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix},$$

with a_3 free. By Theorem 2.5.4, $\text{nul}(A) = \text{span}((2, -1, 1))$. Hence $U^0 = \text{span}(\psi)$ with

$$\psi(x, y, z) = 2x - y + z.$$

Check: $\psi(1, 2, 0) = 2 - 2 + 0 = 0$ and $\psi(0, 1, 1) = -1 + 1 = 0$. The two spanning vectors are independent (neither is a multiple of the other), so $\dim U = 2$, and $\dim U + \dim U^0 = 2 + 1 = 3$, as Theorem 4.2.2 predicts. As a by-product, U is the plane $2x - y + z = 0$: $U \subseteq \ker \psi$, both have dimension 2 (Proposition 4.1.2), and Theorem 1.6.8 gives equality.

The same recipe works in general. For $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$, let A have rows $\mathbf{u}_1^T, \dots, \mathbf{u}_m^T$. Then $U^0 = \{\varphi_{\mathbf{a}} : \mathbf{a} \in \text{nul}(A)\}$, and the isomorphism of Theorem 4.1.1 carries a basis of $\text{nul}(A)$ to a basis of U^0 .

Away from F^n , counting often replaces the computation entirely.

Example. (The Equations of a Root Condition)

Let $U = \{p \in \mathbb{R}[x]_{\leq 2} : p(1) = 0\}$. Find U^0 .

Solution. Let $\varepsilon_1(p) = p(1)$, a linear functional (Example (Four Families of Functionals)). By definition $U = \ker \varepsilon_1$, so $\varepsilon_1 \in U^0$. Since $\varepsilon_1(1) = 1$, $\varepsilon_1 \neq 0$, and Proposition 4.1.2

gives $\dim U = 3 - 1 = 2$. By Theorem 4.2.2, $\dim U^0 = 3 - 2 = 1$. Now $\text{span}(\varepsilon_1)$ is a subspace of U^0 (Proposition 4.2.1) of dimension 1, so $U^0 = \text{span}(\varepsilon_1)$ by Theorem 1.6.8.

We never found a basis of U and never solved an equation. The conclusion is concrete: a functional on $\mathbb{R}[x]_{\leq 2}$ vanishes on every polynomial with a root at 1 **only if** it is a multiple of $p \mapsto p(1)$.

Quick check. Let $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$. What is $\dim U^0$? Name a basis of U^0 without solving a system.

Solution. $\dim U^0 = 3 - 1 = 2$ by Theorem 4.2.2. The functionals $x - y$ and $y - z$ vanish at $(1, 1, 1)$, hence on U , and they are independent: in $a(x - y) + b(y - z) = 0$, the coefficient of x gives $a = 0$, and then $b = 0$. Two independent vectors in a 2-dimensional space form a basis (Theorem 1.6.4).

4.2.3 Sums, intersections and spans

Taking annihilators reverses inclusions: more vectors satisfy fewer common equations. It also exchanges sums with intersections. In $\mathbb{R}^3$, let U be the xy -plane and W the xz -plane. Then $U^0 = \text{span}(z)$ and $W^0 = \text{span}(y)$, writing x, y, z for the coordinate functionals. The intersection $U \cap W$ is the x -axis, with annihilator $\text{span}(y, z) = U^0 + W^0$. The sum $U + W = \mathbb{R}^3$ has annihilator $\{0\} = U^0 \cap W^0$. The theorem says this always happens.

Theorem 4.2.3 — Properties of Annihilators

Let V be a vector space over F .

- (a) If $S \subseteq T \subseteq V$, then $T^0 \subseteq S^0$.
- (b) For every subset $S \subseteq V$, $S^0 = (\text{span } S)^0$.
- (c) For subspaces U, W of V , $(U + W)^0 = U^0 \cap W^0$.
- (d) If V is **finite-dimensional**, then for subspaces U, W of V , $(U \cap W)^0 = U^0 + W^0$.

Idea. Parts (a)–(c) are unwinding definitions: fix a functional in one side and test it on the vectors of the other. In (d), the inclusion $U^0 + W^0 \subseteq (U \cap W)^0$ is also direct. The reverse inclusion is the hard half. Given φ vanishing on $U \cap W$, we would have to **split** it as $\psi + \chi$ with ψ killing U and χ killing W , and nothing in sight tells us how. So we count instead: one inclusion plus equal dimensions gives equality. The plan is ① compute $\dim(U^0 + W^0)$ by the dimension formula for sums inside V^* ; ② replace $U^0 \cap W^0$ by $(U + W)^0$ using (c); ③ turn every annihilator dimension into a dimension in V with Theorem 4.2.2; ④ recognize the dimension formula in V . Finite dimension is spent in step ③.

Proof.

- (a) Let $\varphi \in T^0$. Every $\mathbf{s} \in S$ lies in T , so $\varphi(\mathbf{s}) = 0$. Hence $\varphi \in S^0$.
- (b) Since $S \subseteq \text{span } S$ (Theorem 1.3.1), part (a) gives $(\text{span } S)^0 \subseteq S^0$. Conversely, let $\varphi \in S^0$ and $\mathbf{v} \in \text{span } S$, say $\mathbf{v} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k$ with $\mathbf{s}_i \in S$ and $k \geq 0$. By

Theorem 3.1.4, $\varphi(\mathbf{v}) = \sum_i a_i \varphi(\mathbf{s}_i) = 0$. Hence $\varphi \in (\text{span } S)^0$.

(c) Since $U \subseteq U+W$ and $W \subseteq U+W$ (Theorem 1.7.1), part (a) gives $(U+W)^0 \subseteq U^0$ and $(U+W)^0 \subseteq W^0$, so $(U+W)^0 \subseteq U^0 \cap W^0$. Conversely, let $\varphi \in U^0 \cap W^0$, and let $\mathbf{u} + \mathbf{w} \in U+W$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$. Then $\varphi(\mathbf{u} + \mathbf{w}) = \varphi(\mathbf{u}) + \varphi(\mathbf{w}) = 0 + 0 = 0$. Hence $\varphi \in (U+W)^0$.

(d) $(\supseteq)$ Let $\varphi = \psi + \chi$ with $\psi \in U^0$ and $\chi \in W^0$, and let $\mathbf{v} \in U \cap W$. Since $\mathbf{v} \in U$ and $\mathbf{v} \in W$, $\varphi(\mathbf{v}) = \psi(\mathbf{v}) + \chi(\mathbf{v}) = 0 + 0 = 0$. Hence $U^0 + W^0 \subseteq (U \cap W)^0$.

$(\subseteq)$ Let $n = \dim V$. The subspaces U , W , $U \cap W$ (Theorem 1.2.2) and $U+W$ (Theorem 1.7.1) of V are finite-dimensional by Theorem 1.6.7. The space V^* has dimension n by Corollary 4.1.4, so its subspaces U^0 , W^0 are finite-dimensional too. By the Dimension Formula for Sums (Theorem 1.7.3) in V^* , then (c),

$$\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) = \dim U^0 + \dim W^0 - \dim(U \cap W)^0.$$

By Theorem 4.2.2 applied to U , W and $U+W$, the right-hand side is

$$(n - \dim U) + (n - \dim W) - (n - \dim(U+W)) = n - (\dim U + \dim W - \dim(U+W)) = n - \dim(U \cap W),$$

where the last equality is Theorem 1.7.3 in V . By Theorem 4.2.2 once more, $n - \dim(U \cap W) = \dim(U \cap W)^0$. So $U^0 + W^0$ is a subspace of $(U \cap W)^0$ (by $(\supseteq)$ and Theorem 1.7.1) with the same finite dimension, and Theorem 1.6.8 gives $U^0 + W^0 = (U \cap W)^0$. This proves the theorem. ■

Part (b) is what made the computations of this section legitimate: to find the annihilator of a subspace, test functionals on a spanning list only. Parts (c) and (d) form a dictionary. A subspace given as a **sum** has equations obtained by **intersecting** equation spaces; a subspace given as an **intersection** has equations obtained by **adding** them. Later in this chapter this dictionary becomes a practical method for computing sums and intersections of subspaces.

Remark. Parts (a)–(c) and the inclusion $(\supseteq)$ in (d) hold in every vector space; only the counting step used finite dimension. The equality (d) is in fact true for **every** V , but the proof must then construct the splitting $\varphi = \psi + \chi$ by hand; Exercise C1 below does this. In finite dimension, (a) also becomes strict: if $U \subsetneq W$, then $\dim U < \dim W$ (Theorem 1.6.8), so $\dim W^0 < \dim U^0$ and $W^0 \subsetneq U^0$.

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise A1.

- (a) Define the annihilator S^0 of a subset S of a vector space V .
- (b) True or false: U^0 is a subspace of V . Justify your answer.
- (c) Let U be a 2-dimensional subspace of $\mathbb{R}^5$. What is $\dim U^0$?
- (d) For a finite-dimensional V , what are $\{\mathbf{0}\}^0$ and V^0 ?
- (e) True or false: if $U \subsetneq W$ are subspaces of a finite-dimensional V , then $W^0 \subsetneq U^0$.

Justify your answer.

- (f) To compute U^0 for $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$, which matrix do you row reduce, and why?

Solution.

- (a) $S^0 = \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}$ (Definition 4.2.1).
 (b) False. U^0 is a subspace of the dual space V^* (Proposition 4.2.1); its elements are functionals, not vectors of V .
 (c) $\dim U^0 = 5 - 2 = 3$, by Theorem 4.2.2.
 (d) $\{\mathbf{0}\}^0 = V^*$ and $V^0 = \{0\}$, by Example (First Annihilators) (a).
 (e) True. By Theorem 4.2.3 (a), $W^0 \subseteq U^0$. Since $U \subsetneq W$, $\dim U < \dim W$ by Theorem 1.6.8, so $\dim W^0 = \dim V - \dim W < \dim V - \dim U = \dim U^0$. Hence the inclusion is strict.
 (f) The matrix A whose **rows** are $\mathbf{u}_1^T, \dots, \mathbf{u}_m^T$. By Theorem 4.1.1 a functional is $\mathbf{x} \mapsto \mathbf{a}^T \mathbf{x}$, and it kills every $\mathbf{u}_i$ exactly when $A\mathbf{a} = \mathbf{0}$; so U^0 corresponds to $\text{nul}(A)$ (Example (An Annihilator as a Null Space)).

4.2.4.2 B. Practice

Exercise B1: Computing annihilators.

- (a) Let $U = \text{span}((1, 2, 1, 0), (2, 4, 3, 1), (1, 2, 2, 1)) \subseteq \mathbb{R}^4$. Find a basis of U^0 , and verify that $\dim U + \dim U^0 = 4$.
 (b) Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$. Show that $U^0 = \text{span}(\varepsilon_1, \varepsilon_{-1})$, where $\varepsilon_c(p) = p(c)$.

Solution.

- (a) Let A have the three spanning vectors as rows. As in Example (An Annihilator as a Null Space), $\varphi_{\mathbf{a}} \in U^0$ if and only if $A\mathbf{a} = \mathbf{0}$. Subtracting 2 times row 1 from row 2 and row 1 from row 3 gives rows $(1, 2, 1, 0)$, $(0, 0, 1, 1)$, $(0, 0, 1, 1)$; subtracting row 2 from row 3 and then row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are a_2 and a_4 , and Theorem 2.5.4 gives $\text{nul}(A) = \text{span}((-2, 1, 0, 0), (1, 0, -1, 1))$. Hence U^0 has the basis

$$\psi_1(x_1, x_2, x_3, x_4) = -2x_1 + x_2, \quad \psi_2(x_1, x_2, x_3, x_4) = x_1 - x_3 + x_4.$$

Check on the first two spanning vectors: $\psi_1(1, 2, 1, 0) = 0$, $\psi_2(1, 2, 1, 0) = 0$, $\psi_1(2, 4, 3, 1) = 0$, $\psi_2(2, 4, 3, 1) = 2 - 3 + 1 = 0$; the third vector is their difference, $(2, 4, 3, 1) - (1, 2, 1, 0)$. The RREF has two pivots, so $\dim U = \dim \text{row}(A) = 2$ (Theorem 2.5.5), and $2 + 2 = 4$.

- (b) By definition $\varepsilon_1, \varepsilon_{-1} \in U^0$. *Independent:* if $a\varepsilon_1 + b\varepsilon_{-1} = 0$, applying it to $x + 1$ gives $2a = 0$ and applying it to $x - 1$ gives $-2b = 0$. *Dimension of U :* the map

$T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p(-1))$, is linear with kernel U , and its image contains $T(x+1) = (2, 0)$ and $T(x-1) = (0, -2)$, which span $\mathbb{R}^2$. So $\text{rank } T = 2$ and $\dim U = 4 - 2 = 2$ by Theorem 3.3.1. By Theorem 4.2.2, $\dim U^0 = 2$. Hence the 2-dimensional subspace $\text{span}(\varepsilon_1, \varepsilon_{-1})$ of U^0 is all of U^0 (Theorem 1.6.8).

Exercise B2: Equations of the symmetric matrices.

Let $U = \{A \in M_2(\mathbb{R}) : A^T = A\}$. Find U^0 , and verify the dimension formula.

Solution. Every symmetric matrix is $aE_{11} + b(E_{12} + E_{21}) + cE_{22}$, so $U = \text{span}(E_{11}, E_{12} + E_{21}, E_{22})$. Let (ε_{ij}) be the dual basis of the standard basis, so that every $\psi \in M_2(\mathbb{R})^*$ is $\psi = \sum_{i,j} c_{ij}\varepsilon_{ij}$ with $c_{ij} = \psi(E_{ij})$ (Theorem 4.1.3 (c)). By Theorem 4.2.3 (b), $\psi \in U^0$ if and only if ψ kills the three spanning matrices:

$$c_{11} = 0, \quad c_{12} + c_{21} = 0, \quad c_{22} = 0.$$

So $\psi = c_{12}(\varepsilon_{12} - \varepsilon_{21})$, and $U^0 = \text{span}(\varepsilon_{12} - \varepsilon_{21})$: the only equation of the symmetric matrices, up to scaling, is $a_{12} - a_{21} = 0$. The three spanning matrices are independent (they involve disjoint sets of entries), so $\dim U = 3$, and $\dim U + \dim U^0 = 3 + 1 = 4 = \dim M_2(\mathbb{R})$.

Exercise B3: The equations of a hyperplane.

Let V be a finite-dimensional vector space over F , and let $\varphi \in V^*$ be non-zero. Prove that $(\ker \varphi)^0 = \text{span}(\varphi)$.

Solution. By definition of the kernel, φ vanishes on $\ker \varphi$, so $\varphi \in (\ker \varphi)^0$, and $\text{span}(\varphi) \subseteq (\ker \varphi)^0$ by Proposition 4.2.1. Since $\varphi \neq 0$, $\dim \text{span}(\varphi) = 1$, and Proposition 4.1.2 gives $\dim \ker \varphi = \dim V - 1$. By Theorem 4.2.2, $\dim(\ker \varphi)^0 = \dim V - (\dim V - 1) = 1$. A 1-dimensional subspace of a 1-dimensional space is the whole space (Theorem 1.6.8), so $(\ker \varphi)^0 = \text{span}(\varphi)$.

4.2.4.3 C. Going deeper

Exercise C1: The intersection formula without counting.

Let U, W be subspaces of a vector space V over F , and let $\varphi \in (U \cap W)^0$.

- Suppose V is finite-dimensional. Construct $\psi \in U^0$ and $\chi \in W^0$ with $\varphi = \psi + \chi$, without using any dimension formula.
- Explain how to modify the construction when V is infinite-dimensional. Deduce that $(U \cap W)^0 = U^0 + W^0$ holds in every vector space.

Hint: extend a basis of $U \cap W$ to a basis of U and, separately, to a basis of W .

Solution.

- Let $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ be a basis of $U \cap W$ (Theorem 1.2.2, Theorem 1.6.7). Extend it to a basis $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p)$ of U and to a basis $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{w}_1, \dots, \mathbf{w}_q)$ of W

W (Theorem 1.6.5).

The combined list $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_1, \dots, \mathbf{w}_q)$ is independent. Let $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i + \sum c_j \mathbf{w}_j = \mathbf{0}$. Then $\sum c_j \mathbf{w}_j = -\sum a_i \mathbf{x}_i - \sum b_i \mathbf{u}_i$ lies in W and in U , hence in $U \cap W$, so $\sum c_j \mathbf{w}_j = \sum d_i \mathbf{x}_i$ for some $d_i \in F$. Since $(\mathbf{x}, \mathbf{w})$ is a basis of W , all c_j and d_i are 0. Then $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i = \mathbf{0}$, and since $(\mathbf{x}, \mathbf{u})$ is a basis of U , all a_i and b_i are 0.

Extend the combined list to a basis of V by vectors $\mathbf{y}_1, \dots, \mathbf{y}_r$ (Theorem 1.6.5). By Theorem 3.1.5 there is $\psi \in V^*$ with

$$\psi(\mathbf{x}_i) = 0, \quad \psi(\mathbf{u}_i) = 0, \quad \psi(\mathbf{w}_j) = \varphi(\mathbf{w}_j), \quad \psi(\mathbf{y}_l) = 0.$$

Then ψ kills a basis of U , so $\psi \in U^0$ (Theorem 4.2.3 (b)). Put $\chi := \varphi - \psi$. Then $\chi(\mathbf{x}_i) = \varphi(\mathbf{x}_i) - 0 = 0$, because $\mathbf{x}_i \in U \cap W$ and $\varphi \in (U \cap W)^0$; and $\chi(\mathbf{w}_j) = \varphi(\mathbf{w}_j) - \varphi(\mathbf{w}_j) = 0$. So χ kills a basis of W , and $\chi \in W^0$. Hence $\varphi = \psi + \chi \in U^0 + W^0$.

- (b) Replace the finite lists by sets. By Theorem 1.8.3, $U \cap W$ has a basis X , and by Theorem 1.8.2 it extends to a basis $X \cup P$ of U and to a basis $X \cup Q$ of W , with $P \cap X = Q \cap X = \emptyset$. First, $P \cap Q = \emptyset$: a vector of $P \cap Q$ lies in $U \cap W = \text{span } X$, which contradicts the independence of $X \cup P$. So every vector of $X \cup P \cup Q$ lies in exactly one of X , P , Q . A linear relation among finitely many distinct vectors of $X \cup P \cup Q$ splits into an X -part, a P -part and a Q -part, and the independence argument of (a) applies word for word; so $X \cup P \cup Q$ is independent. By Theorem 1.8.2 it extends to a basis B of V . Every $\mathbf{v} \in V$ is a finite combination $\mathbf{v} = \sum_{\mathbf{b}} c_{\mathbf{b}} \mathbf{b}$ of distinct vectors of B , with unique coefficients because B is independent. Define

$$\psi(\mathbf{v}) := \sum_{\mathbf{b} \in Q} c_{\mathbf{b}} \varphi(\mathbf{b}),$$

a finite sum over the vectors of Q that occur. Adding or scaling expansions gives expansions, so by uniqueness the coefficients add and scale, and ψ is linear. It vanishes on $X \cup P$ and agrees with φ on Q , so the rest of (a) is unchanged: $\psi \in U^0$ and $\chi = \varphi - \psi$ kills the basis $X \cup Q$ of W . Together with the inclusion ($\supseteq$) of Theorem 4.2.3 (d), which used no finiteness, this gives $(U \cap W)^0 = U^0 + W^0$ in every vector space.

Exercise C2: Equations of a double root.

Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0 \text{ and } p'(1) = 0\}$, and let $\varepsilon_1(p) = p(1)$ and $\eta_1(p) = p'(1)$. Prove that $U^0 = \text{span}(\varepsilon_1, \eta_1)$. Deduce that $\psi(p) = p(0) - p(2) + 2p'(1)$ does **not** vanish on U .

Solution. Both ε_1 and η_1 are linear (evaluation, and differentiation followed by evaluation), and they vanish on U by definition. *Independent:* if $a\varepsilon_1 + b\eta_1 = 0$, applying it to 1 gives $a = 0$, and applying it to $x - 1$ gives $b = 0$. *Dimension of U :* the linear map $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p'(1))$, has kernel U , and $T(1) = (1, 0)$, $T(x - 1) = (0, 1)$, so T is surjective. By Theorem 3.3.1, $\dim U = 4 - 2 = 2$, and by Theorem 4.2.2, $\dim U^0 = 2$. The 2-dimensional subspace $\text{span}(\varepsilon_1, \eta_1)$ of U^0 is therefore all of U^0 (Theorem 1.6.8).

For the deduction, suppose $\psi \in U^0$. Then $\psi = a\varepsilon_1 + b\eta_1$. Applying both sides to 1 gives $1 - 1 + 0 = a$, so $a = 0$; applying them to $x - 1$ gives $-1 - 1 + 2 = b$, so

$b = 0$. Then $\psi = 0$. But $\psi(x^2) = 0 - 4 + 4 = 0$ and $\psi(x^3) = 0 - 8 + 6 = -2 \neq 0$, a contradiction. Hence $\psi \notin U^0$. Check with an explicit witness: $p = x(x-1)^2 \in U$ has $\psi(p) = 0 - 2 + 0 = -2 \neq 0$.

Exercise C3: Dual of a direct sum decomposition.

- (a) Let V be finite-dimensional with $V = U \oplus W$. Prove that $V^* = U^0 \oplus W^0$.
- (b) In $\mathbb{R}^2$, let $U = \text{span}((1,0))$ and $W = \text{span}((1,1))$. Write $\psi(x,y) = x$ as $\psi_U + \psi_W$ with $\psi_U \in U^0$ and $\psi_W \in W^0$.

Solution.

- (a) Since the sum is direct, $U \cap W = \{\mathbf{0}\}$ (Theorem 1.7.4), and $U + W = V$. By Theorem 4.2.3 (c) and Example (First Annihilators) (a), $U^0 \cap W^0 = (U + W)^0 = V^0 = \{\mathbf{0}\}$. By Theorem 4.2.3 (d), $U^0 + W^0 = (U \cap W)^0 = \{\mathbf{0}\}^0 = V^*$. So $V^* = U^0 + W^0$, and the sum is direct by Theorem 1.7.4.
- (b) A functional $ax + by$ vanishes on $(1,0)$ if and only if $a = 0$, so $U^0 = \text{span}(y)$. It vanishes on $(1,1)$ if and only if $b = -a$, so $W^0 = \text{span}(x - y)$. Solving $x = sy + t(x - y)$ gives $t = 1$ and $s - t = 0$, so $s = 1$:

$$\psi = \underbrace{y}_{\in U^0} + \underbrace{(x - y)}_{\in W^0}.$$

By (a) this decomposition is unique.